Secure Monitoring Platform Requirements

Software Requirements Specification

IEEE 830-1998 outline

DRAFT — human review required

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1 Introduction

1.1 Purpose

This specification records the project purpose, stakeholder constraints and unresolved decisions stated in the source evidence below.

This SRS describes the software's required behavior for users, administrators and the Project Analyst reviewing the specification.

1.2 Scope

Design a secure monitoring platform for university servers and managed endpoints that collects system events, authentication activity, process information, and security alerts from multiple departments. The platform should detect unusual login behavior, repeated failed authentication, suspicious process activity, and unauthorized configuration changes. Security administrators should be able to review alerts, investigate affected devices, record investigation notes, change alert status, and isolate compromised endpoints when necessary. The system should maintain audit logs of security events and administrator actions and provide a centralized dashboard for monitoring security status across multiple campus locations.

1.3 Definitions, acronyms and abbreviations

Term	Meaning
Device isolation	Restriction of a device's network access; the source determines whether and by whom this action is permitted.
Log	A collection of event records. This term does not itself require additional logging fields, events or access permissions.
Audit record	A stored record of an event. Required event scope and retention are determined by the linked source, not this definition.

1.4 References

Reference	Source
Format	IEEE Std 830-1998, clauses 5 and Annex A (document organization).

1.5 Overview

Section 2 summarizes the product, stakeholders and open decisions. Section 3 contains the uniquely identified requirements. Detailed evidence, acceptance procedures and the SDLC plan are available in the supporting report.

2 Overall description

2.1 Product perspective

Design a secure monitoring platform for university servers and managed endpoints that collects system events, authentication activity, process information, and security alerts from multiple departments.

2.2 Product functions

Functional topics are summarized below; section 3 retains the full requirements.

- Authentication — FR-001, FR-002
- Monitoring and alerts — FR-001, FR-003
- Event investigation — FR-003
- Network isolation — FR-003
- Device support — FR-003

2.3 User characteristics and stakeholders

Labels group stakeholders for readability. They do not combine permissions.

Education, training and experience requirements remain unspecified.

Role	Source user groups	Responsibility
Admin	administrators	Performs only the role's source-defined operations in section 3.
Project Analyst	Document review role	Reviews requirements and tracks open decisions; this role grants no product access.

2.4 Constraints

No additional implementation constraints were confirmed. Source-defined restrictions remain in the requirements in section 3.

2.5 Assumptions and dependencies

Open decisions requiring confirmation:

- Data feed encryption method
- Event normalization schema
- Threshold configuration values
- Baseline data collection frequency
- Endpoint isolation workflow
- Access control for investigation actions
- The specific data fields included in audit logs
- The granularity of security status indicators on the dashboard
- Whether real-time updates are provided to the dashboard
- Which specific data fields should be included in audit logs?
- What is the granularity of security status indicators on the centralized dashboard?
- Should real-time updates be provided to the dashboard?
- How will endpoint isolation workflows be defined and executed?
- What access controls will govern investigation actions during incident response?
- What encryption method will be used for data feeds between departments and servers?
- What schema will be used to normalize collected events from multiple sources?

- What threshold configuration values will trigger alerts for unusual behavior?
- What baseline data collection frequency will be maintained for behavioral analysis?

3 Specific requirements

Each requirement is stated once with its original identifier and wording. Cross-references identify requirements covering more than one concern.

3.1 External interface requirements

3.1.1 User interfaces

Related requirements: NFR-001. Interface details beyond those requirements remain unspecified.

3.1.2 Hardware interfaces

Related requirements: FR-003. Interface details beyond those requirements remain unspecified.

3.1.3 Software interfaces

Related requirements: FR-001, FR-002. Interface details beyond those requirements remain unspecified.

3.1.4 Communications interfaces

No specific interface requirement was supplied; confirmation is required.

3.2 Functional requirements

FR-001 · Must · explicit

The system shall collect system events, authentication activity, process information, and security alerts from multiple departments via secure data feeds.

Traceability: E-001, TEST-001

FR-002 · Should · explicit

The system shall detect unusual login behavior, repeated failed authentication, suspicious process activity, and unauthorized configuration changes using predefined behavioral thresholds.

Traceability: E-002, TEST-002

FR-003 · Should · explicit

Security administrators shall be able to review alerts, investigate affected devices, record investigation notes, change alert status, and isolate compromised endpoints when necessary.

Traceability: E-003, TEST-003

3.3 Performance requirements

No separate requirement was specified in the supplied brief.

3.4 Logical database requirements

No separate requirement was specified in the supplied brief.

3.5 Design constraints

No separate requirement was specified in the supplied brief.

3.6 Software system attributes

3.6.1 Reliability

No separate requirement was specified in the supplied brief.

3.6.2 Availability

No separate requirement was specified in the supplied brief.

3.6.3 Security

Also covered by: FR-001, FR-002, FR-003.

NFR-001 · Should · explicit

The system shall maintain audit logs of security events and administrator actions and provide a centralized dashboard for monitoring security status across multiple campus locations.

Traceability: E-004, TEST-004

3.6.4 Maintainability

No separate requirement was specified in the supplied brief.

3.6.5 Portability

No separate requirement was specified in the supplied brief.

3.7 Other requirements

No separate requirement was specified in the supplied brief.

Detailed acceptance criteria, evidence, architecture and the SDLC plan are available in the supporting report and canonical JSON.

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